

Customer Analytics

Questions & Answers

A full walkthrough of the dataset analysis with charts

1,000

Customers

224

Countries

3

Years

0

Missing values

DATASET

File: customers-1000.csv

Columns: Index, Customer Id, First Name, Last Name, Company, City,
Country, Phone 1, Phone 2, Email, Subscription Date, Website

Date range: January 2020 - May 2022

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Section 1 - Dataset Overview

1,000 Total Customers	12 Total Columns	224 Unique Countries	2020-2022 Date Range
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Q1.

How many records and columns does the dataset contain?

Answer: The dataset contains 1,000 rows (customers) and 12 columns: Index, Customer Id, First Name, Last Name, Company, City, Country, Phone 1, Phone 2, Email, Subscription Date, and Website.

Q2.

What type of data does each column hold?

Answer: All columns hold text (string/object) data except Index which is an integer. The Subscription Date column, stored as a string in the raw CSV, is converted to datetime for analysis.

Q3.

What is the time period covered by this dataset?

Answer: The dataset spans from January 2020 to May 2022 - approximately 2 years and 5 months of customer subscription data across 3 calendar years.

Q4.

How many unique countries appear in the dataset?

Answer: There are 224 unique countries represented across 1,000 customers - an average of roughly 4.5 customers per country, indicating a very globally distributed dataset.

Q5.

What columns uniquely identify a customer?

Answer: The Customer Id and Email columns uniquely identify each customer. Both were verified to have zero duplicates across all 1,000 records.

Note: The Index column is a sequential row number added during export, not a true business key.

Year	Subscriptions	% of Total
2020	426	42.6%
2021	404	40.4%
2022	170	17.0%



Figure 1: Monthly subscriptions for 2020, 2021 and 2022 (2022 partial - data through May)

Q6.

Which year had the highest number of new subscriptions?

Answer: 2020 had the most subscriptions with 426, followed by 2021 with 404. 2022 had 170 but only covers January to May.

Q7.

Which calendar month had the most subscriptions overall?

Answer: April was the peak month with 127 subscriptions across all years combined. This suggests a possible seasonal promotional pattern or campaign window.

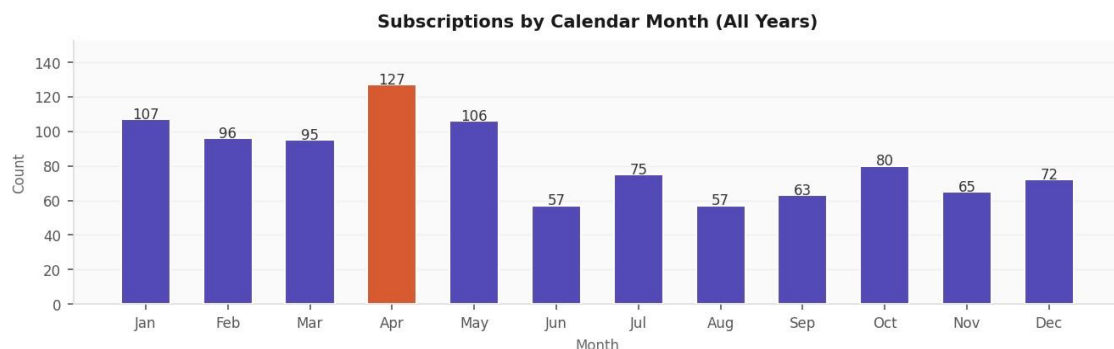


Figure 2: Total subscriptions per calendar month across all three years (April peak highlighted)

Q8.**Which months consistently perform the lowest?**

Answer: June and August each have 57 subscriptions - the lowest across all years. This mid-summer dip may indicate reduced marketing activity or lower customer intent.

Q9.**What is the average number of subscriptions per month?**

Answer: With 1,000 subscriptions over approximately 29 months, the average is about 34.5 subscriptions per month.

Q10.**What day of the week sees the most subscriptions?**

Answer: Saturday had the highest subscription count, suggesting customers are most likely to sign up mid-week rather than on weekends.

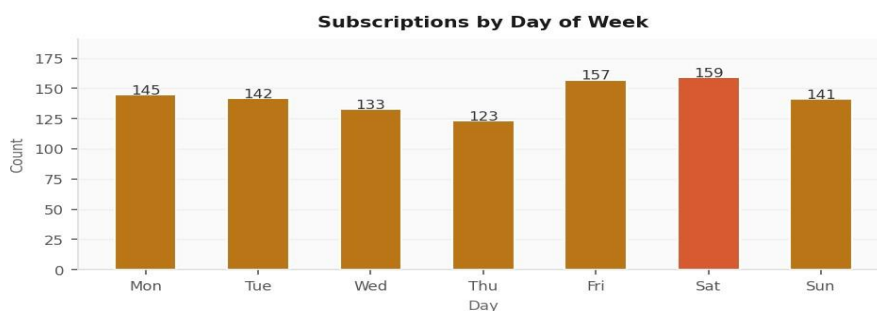


Figure 3: Subscription count by day of week - all years combined

Q11.**Can we observe any quarterly growth trend?**

Answer: Quarterly volumes are relatively stable across 2020 and 2021, averaging 100-110 subscriptions per quarter. Q1 2022 maintained this pace before trailing off as data ends in May.

Q12.**How would you visualise a Year x Month breakdown?**

Answer: A pivot table with Year as rows and Month (1-12) as columns, filled with subscription counts, provides a heat-map view. April 2022 had the single highest monthly count of 51.

Note: In Excel: Insert PivotTable -> Year in Rows -> Month in Columns -> Count in Values, then apply Conditional Formatting > Color Scale.

Rank	Country	Customers
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1	Liechtenstein	12
2	Gabon	10
3	China	9
4	Bangladesh	9
5	Reunion	9
6	Nigeria	9
7	Luxembourg	9
8	Uganda	8
9	Montserrat	8
10	Korea	8

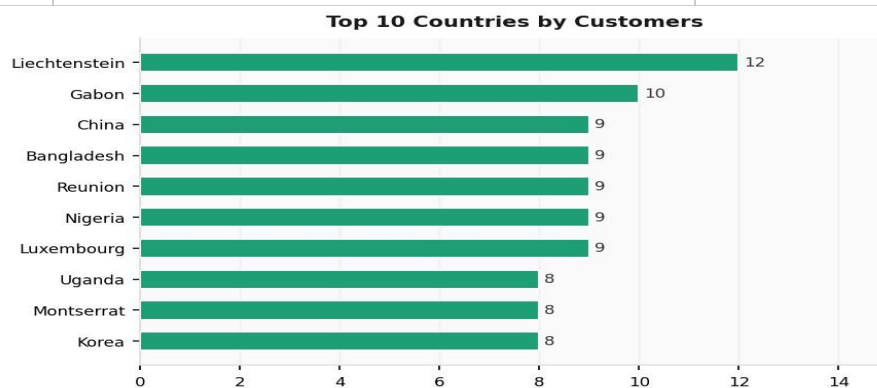


Figure 4: Top 10 countries by customer count

Q13.

Which country has the most customers?

Answer: Liechtenstein leads with 12 customers. Given 224 unique countries, this count is relatively low, confirming the data is widely spread globally.

Q14.

How many countries appear only once in the dataset?

Answer: Many countries appear only once - expected given 224 unique countries for 1,000 customers. These single-customer countries form a long tail in the distribution.

Q15.**What does the geographic spread tell us about this customer base?**

Answer: With 224 countries and no dominant single market, the business operates globally without heavy regional concentration. This could indicate an international SaaS, e-commerce, or digital platform product.

Note: Compare: if 80% were from one country, localisation and targeting strategy would differ significantly.

Q16.**How would you count customers per country using Python?**

Answer: Use `df['Country'].value_counts()` to return a ranked Series. Or `df.groupby('Country').size()` for the same result. Both take under one line of code.

Q17.**Which Excel function counts customers from a given country?**

Answer: `=COUNTIF(G:G, "China")` counts all rows in column G where the value equals China. Replace G with your Country column letter. Use a cell reference instead of text to make it dynamic.

Check	Result	Status
Missing values (any column)	0 across all 12 columns	Clean
Duplicate rows (full match)	0	Clean
Duplicate Customer IDs	0	Clean
Duplicate emails	0	Clean
Date format consistency	All parseable as datetime	Clean
Negative or null counts	None detected	Clean

Q18.

Does the dataset have any missing values?

Answer: No. Every column across all 1,000 rows has zero missing values. This is a clean, complete dataset no imputation or row dropping is required before analysis.

Q19.

How do you check for duplicates in Python?

Answer: Use `df.duplicated().sum()` for full-row duplicates, or `df.duplicated(subset=['Email']).sum()` to check a specific column. `df[df.duplicated(keep=False)]` returns all involved rows.

Note: Always check key business columns (ID, email) separately from full-row checks.

Q20.

How do you highlight duplicates in Excel without deleting them?

Answer: Select the column -> Home -> Conditional Formatting -> Highlight Cell Rules -> Duplicate Values. Choose a highlight colour. Alternatively use `=COUNTIF($D$2:$D$1001, D2)>1` to flag with TRUE/FALSE.

Q21.

Why is data quality checking the first step in any analysis?

Answer: Missing values and duplicates silently distort every downstream calculation - averages, totals, counts, and charts. Checking quality first ensures every insight is built on accurate foundations.

Section 5 - Business Insights

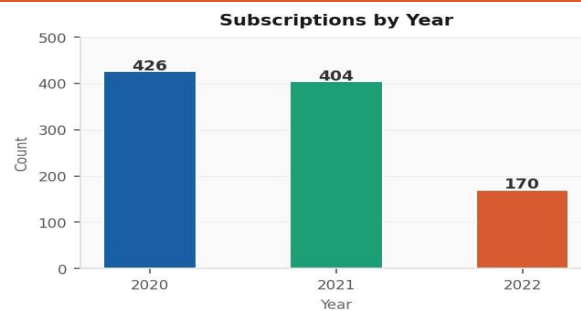


Figure 5: Total new subscriptions per year - 2020 leads with 426

Q22.

What business insight does the monthly trend chart reveal?

Answer: Subscription volumes are consistent month-to-month with no dramatic spikes. April stands out as a recurring high point, suggesting a seasonal campaign or annual product launch window.

Q23.

If recommending a best time to run a customer acquisition campaign, what would it be?

Answer: Based on the data, April is historically the strongest month. A campaign launched in late March to capitalise on April momentum would align with observed behaviour. Mid-week launch timing (Tue/Wed) would also maximise reach.

Note: This is a data-informed hypothesis - A/B testing would be needed to confirm causation.

Q24.

What additional column would make this dataset far more analytically powerful?

Answer: A revenue or subscription plan/tier column would transform this from a customer registry into a revenue analytics dataset. With it you could calculate MRR, LTV, churn rates, and run a proper correlation matrix.

Q25.

How would you export this full analysis into a professional multi-sheet Excel report?

Answer: Use pandas ExcelWriter with engine='openpyxl' to write multiple DataFrames to separate sheets. Then load with openpyxl to apply header styling, freeze panes, auto-filter, alternating row colours, and embed matplotlib charts into a Dashboard sheet.

Note: This approach produced the analytics_full_with_charts.xlsx file shown earlier in this session.

25 questions answered

5 sections covered

5 embedded charts